

To The Institute for Global Environmental Strategies

ENVIRONMENTAL POLLUTION IN VIET NAM

Report

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1. Vietnam industrial wastewater standards

			Vietnam industrial wastewater standards				Japan uniform drainage standard (Ministry of the Environment)	
			Vietnam standard (encouraging compliance)		National Technical Regulations (Regulatory Basis)			
			TCVN5945:2010		QCVN 40:2011			
			A	B	A	B		
No	Parameter	Unit						
1	Temperature	°C	40	40	40	40	-	
-	Odor		Not uncomf ortable	Not uncomf ortable	-	-	-	
2	Color	Co-Pt	20	70	50	150	-	
	Identify	Correct	Co-Pt2, pH7		Co-Pt, pH7			
3	pH	-	6 - 9	5.5 - 9	6 - 9	5.5 – 9	5.8 - 8.6 (river) 5.0 - 9.0 (marine area)	
4	BOD ₅ (20°C)	mg/l	30	50	30	50	160 Day average 120	
5	COD Cr method	mg/l	50	100	75	150	160 Day average 120	COD _{Mn}
6	TSS	mg/l	50	100	50	100	200 Day average 150	
7	As	mg/l	0.05	0.1	0.05	0.1	0.1	

Analysis
QCVN 40: 2011 (Japanese translation reference QCVN 24:2009)
TCVN 4557: 1988 Water quality - Method for dertermination of temperature
TCVN 6185:2015 Water quality - Examination and determination of color
TCVN 6492: 2011 (ISO 10523: 2008) Water quality -- Determination of pH
TCVN 6001 - 1: 2008 (ISO 5815 - 1: 2003) Water quality - Determination of biochemical oxygen demand after n days (BODn) -- Part 1: Dilution and seeding method with allylthiourea addition
TCVN 6491: 1999 (ISO 6060: 1989) Water quality - Determination of the chemical oxygen demand
TCVN 6625: 2000 (ISO 11923: 1997) Water quality - Determination of suspended solids by filtration through glass-fibre filters
TCVN 6626: 2000 Water quality - Determination arsenic - Atomic absorption spectrometric method (hydride technique)

8	□ Hg	mg/l	0.005	0.01	0.005	0.01	0.005	Mercury and alkyl mercury and other mercury compounds	TCVN 7877: 2008 (ISO 5666-1999) Water quality – Determination of mercury
							Not to be detected	Alkyl mercury compound	
9	Pb	mg/l	0.1	0.5	0.1	0.5	0.1		TCVN 6193: 1996 Water quality – Determination of cobalt nickel, copper, zinc, cadmium, and lead. Flame atomic absorption spectrometric methods
10	Cd	mg/l	0.1	0.2	0.05	0.1	0.03		TCVN 6193: 1996 Water quality – Determination of cobalt nickel, copper, zinc, cadmium, and lead. Flame atomic absorption spectrometric methods
11	Cr ⁶⁺	mg/l	0.05	0.1	0.05	0.1	0.5	mg-Cr(VI)/L	TCVN 6658: 2000 (JSA ISO 11083: 1994) Water quality – Determination of chromium (VI). Spectrometric method using 1,5-diphenylcarbazide
12	Cr ³⁺	mg/l	0.2	1	0.2	1	-		TCVN 6222: 2008 Water quality- Determination of chromium - Atomic absorption spectrometric method
-							2	Cr Content (mg/L)	
13	Cu	mg/l	2	2	2	2	3		TCVN 6193: 1996 Water quality – Determination of cobalt nickel, copper, zinc, cadmium, and lead. Flame atomic absorption spectrometric methods
14	Zn	mg/l	3	3	3	3	2		TCVN 6193: 1996 Water quality – Determination of cobalt nickel, copper, zinc, cadmium, and lead. Flame atomic absorption spectrometric methods
15	Ni	mg/l	0.2	0.5	0.2	0.5	-		TCVN 6193: 1996 Water quality – Determination of cobalt nickel, copper, zinc, cadmium, and lead. Flame atomic absorption spectrometric methods
16	Mn	mg/l	-	-	0.5	1	10	Soluble manganese	TCVN 6002: 1995 Water quality – Determination of manganese - Formaldoxime spectrometric method
17	Fe	mg/l	-	-	1	5	10	Solubility of iron	TCVN 6177: 1996 (ISO 6332-1988) Water quality – Determination of iron - Spectrometric method using 1,10-phenantrolin TCVN 6665: 2011 (ISO 11885:

									2007) Water quality -- Determination of selected elements by inductively coupled plasma optical emission spectrometry (ICP-OES)
-	Tin	mg/l	-	-	-	-	-		
18	Total cyanide	mg/l	0.07	0.1	0.07	0.1	1	mg-CN/L	TCVN 6181: 1996 (ISO 6703-1- 1984) Water quality - Determination of total cyanide
19	Total phenol	mg/l	0.1	0.5	0.1	0.5	5		TCVN 6216: 1996 (ISO 6439- 1990) Water quality -- Determination of phenol index - - 4-Aminoantipyrine spectrometric methods after distillation TCVN 6199-1: 1995 (ISO 8165/1: 1992) Water quality -- Determination of selected monovalent phenols -- Part 1: Gas-chromatographic method after enrichment by extraction
20	Mineral oil	mg/l	1	5	5	10	5	Normal hexane extract substance content (mineral oil content)	TCVN 5070: 1995 Water quality - Weight method for determination of oil and oil product
-	Fats and oils (animal and vegetable oils)	mg/l	-	-	-	-	30	Normal hexane extract substance content (animal and vegetable fat and oil content)	TCVN 7875: 2008 Water - Determination of oil and grease - Partition-infrared method United States Environmental Protection Agency (EPA) Analysis Method 1664 Measurement of total fats and oils based on extracted weight method (fats and oils and total petroleum hydrocarbons)
21	Sulfur compound	mg/l	0.2	0.5	0.2	0.5	-		TCVN 6637: 2000 (ISO 10530: 1992) Water quality - Determination of dissolved sulfide - Photometric method using methylene blue
22	Fluoride	mg/l	-	-	5	10	8 (public waters) 16 (Sea)	Fluorine and its compounds	TCVN 6494-1: 2011 (ISO 10304-1: 2007) Water quality -- Determination of dissolved anions by liquid chromatography of ions -- Part 1: Determination of bromide, chloride, fluoride, nitrate, nitrite, phosphate and sulfate
23	NH ₄ ⁺ -N	mg/l	5	10	5	10	-	100mg / l (NH ₄ ⁺ - N × 0.4 + NO ₂ ⁻ - N + NO ₃ ⁻ - N)	TCVN 5988: 1995 (ISO 5664: 1984) Water quality -- Determination of ammonium -- Distillation and titration method TCVN 6620: 2000 Water quality - Determination of ammonium - Potentiometric method

24	T-N	mg/l	15	30	20	40	120 Day average 60		TCVN 6638: 2000 Water quality - Determination of nitrogen - Catalytic digestion after reduction with devarda's alloy
25	T-P	mg/l	4	6	4	6	16 Day average 8		TCVN 6202: 2008 (ISO 6878: 2004) Water quality -- Determination of phosphorus -- Ammonium molybdate spectrometric method
26	Chloride	mg/l	500	600	500	1000	-		TCVN 6494-1: 2011 (ISO 10304-1: 2007) Water quality -- Determination of dissolved anions by liquid chromatography of ions -- Part 1: Determination of bromide, chloride, fluoride, nitrate, nitrite, phosphate and sulfate
			Flowing into rivers, streams, lakes that have lower chloride concentrations		not applicable when discharging into saline water and brackish water				
27	Residual chlorine	mg/l	1	2	1	2	-		TCVN 6225-3: 2011 (ISO 7393-3: 1990) Water quality - Determination of free chlorine and total chlorine - Part 3: Iodometric titration method for the determination of total chlorine
28	Pesticide: Organic chlorine	mg/l	undetected	undetected	0.05	0.1	Specified for each substance		TCVN 7876: 2008 Water - Determination of organochlorine pesticides content. Liquid-liquid extraction gas chromatographic method
29	Pesticide: Organic phosphorus	mg/l	undetected	undetected	0.3	1	1	As an organophosphorus compound	TCVN 8062: 2009 Organophosphorus compounds by gas chromatography - Capillary column technique
30	T-PCB	mg/l	-	-	0.003	0.01	0.003		
31	Coliform	MPN/100ml	3000	5000	3000	5000	3000 / cm ³	Day average	TCVN 8775: 2011 - Water quality - Total coliform - Membrane-filter technique TCVN 6187 - 1: 2009 (ISO 9308 - 1: 2000) Water quality - Detection and enumeration of Escherichia coli and coliform bacteria - Part 1: Membrane filtration method TCVN 6187-2: 1996 (ISO 9308-2: 1990 (E)) Water quality - Detection and enumeration of coliform organisms, thermotolerant coliform organisms, and presumptive Escherichia coli - Part 2: Multiple tube (most probable number) method

32	Gross α activity	Bq/l	0.1	0.1	0.1	0.1	-		TCVN 6053: 2011 Water quality - Measurement of gross alpha activity in non-saline water - Thick source method
33	Gross β activity	Bq/l	1	1	1	1	-		TCVN 6219:2011 Water quality - Measurement of gross beta activity in non-saline water - Thick source method
-	Organic insecticide	mg/l	-	-	-	-	-		
-	Organochlorine insecticide	mg/l	-	-	-	-	-		
-	Measurement by biological indicator	-	-	-	-	-	-		
34	Toxicity test with Vibrio Fischeri (30 min) TU	mg/l	TU (Toxicity Units) 8	-	-	-	-		TCVN 6831-1:2010 (ISO 11348-1:2007):Water quality - Determination of the inhibitory effect of water samples on the light emission of Vibrio fischeri (Luminescent bacteria test) - Part 1: Method using freshly prepared bacteria TCVN 6831-2:2010 (ISO 11348-2:2007):Water quality-Determination of the inhibitory effect of water samples on the light emission of Vibrio fischeri (Luminescent bacteria test)- Part 2 : Method using liquid-dried bacteria TCVN 6831-3:2010 (ISO 11348-3:2007): Water quality – Determination of the inhibitory effect of water samples on the light emission of Vibrio fischeri (Luminescent bacteria test) – Part 3: Method using freeze – dried bacteria

QCVN

A: Value of pollutants in industrial wastewater discharged to water source that is used for supplying domestic water

B: Contaminant value in industrial wastewater discharged to water source that isn't used for supplying domestic water

TCVN

A standard or less: It can be discharged to the water source that is constantly used for supplying domestic water

B standard or less: : It can be discharged to the water source that is different from A column

Other

TCVN 6663 - 1: 2011 (ISO 5667 - 1: 2006) Water quality -- Sampling -- Part 1: Guidance on the design of sampling programmes and sampling techniques

TCVN 6663-3: 2008 (ISO 5667-3: 2003) Water quality - Sampling. Guidance on the preservation and handling of water samples

TCVN 5999: 1995 (ISO 5667-10: 1992) Water quality - Sampling - Part 10: Guidance on sampling of wastewaters

Content	TCVN	QCVN
Relation	TCVN are based on: 1) International, regional and foreign standards; 2) Scientific and technological research results and technical advances; 3) Practical experience; 4) Results of assessment, testing, inspection and checking	QCVN are based on: 1) TCVN; 2) International, regional and foreign standards; 3) Achievement of science-technology and technical advance; 4) Results of assessment, testing, inspection and checking
Revision history	1. TCVN 5945:2005 was revised by MOST to refer TCVN 5945:1995 2. TCVN 5945:2010 was revised by improving TCVN 5945:2005	1. QCVN 24:2009 was revised by MONRE to refer TCVN 5945:2005 2. QCVN 40:2011 was revised by improving QCVN 24:2009
Status	TCVN 5945:1995: Withdrawn TCVN 5945:2005: Withdrawn TCVN 5945:2010: Active	QCVN 24:2009: Withdrawn QCVN 40:2011: Active

Note:

TCVN: National Standard

QCVN: National Technical Regulation

MOST: Ministry of Science and Technology

MONRE: Ministry of Natural Resource and Environment

Reference

About TCVN 5945: 2005

URL:<https://www.env.go.jp/air/tech/ine/asia/vietnam/files/needs/needs.pdf>

About QCVN 24: 2009

URL:<https://www.env.go.jp/air/tech/ine/asia/vietnam/files/pollution/pollution.pdf>

QCVN 24: 2009 (Japanese translation)

URL:<https://www.env.go.jp/air/tech/ine/asia/vietnam/files/law/V28.pdf>

2. Pollution Cases in Viet Nam

2.1. Vedan Bien Hoa pollution

2.1.1 Case progress

13th, September, 2008 From feedback and pressing of the local people about the status of clandestine discharge of untreated waste water into the environment, after more than 3 months of follow-up, inter-agency inspection team caught Vedan stationed in Long Thanh district, Dong Nai province discharged a large amount of waste water untreated into the Thi Vai river. Vedan discharging waste into the Thi Vai River is supposed environmental pollutants and to be detected by Environmental Police Department - Vietnam Ministry of Public Security cooperated with the Ministry of Natural Resources and Environment of Vietnam. According to the original judgment, Vedan's discharge system installation violates of the provisions of the law on environmental protection. According to Colonel Luong Minh Thao, violations are extremely serious. At the scene, Deputy Director of the office Vedan Vietnam has admitted violations of the company.

19th, September, 2008, the Ministry of Natural Resources and Environment announced the results of the investigation of Vedan 10 violations, including:

1. Discharge of wastewater exceeds permitted standards by more 10 times for the MSG and lysine factory
2. Discharge of wastewater exceeds permitted standards by more 10 times for modified starch factory
3. Discharge of wastewater exceeds permitted standards by more 10 times for the company's other plants.
4. Submitting incomplete survey data, survey, monitoring and other relevant documents to the agency data storage environmental information as prescribed
5. Didn't register environmental protection commitments with the state authorities on environmental protection for pig farms.
6. No reports of environmental impact assessment which was built and put into operation projects for investment projects to raise capacity base - acid factories from 3,116 tons / month to 6,600 tons / month.

7. Do not report environmental impact assessment which was built and put into operation projects for investment projects to raise capacity of monosodium glutamate factory from 5,000 tons / month to 15,000 tons / month, starch denaturation of 2,000 tons / month to 4,000 tons / month, lysine from 1,200 tons / month to 1,400 tons / month, senior spice powder 20 tons / month, PGA 700 tons / year, 70,000 tons of fertilizer Combination / year (solid), 280,000 tons / year (liquid).
8. Emission stench, unpleasant odors directly into the environment without treatment
9. Management of hazardous wastes contravenes regulations about environmental protection.
10. Companies discharging waste water into water sources incorrectly specified location in the permit.

2.1.2. Solutions

-6th, October, 2008 Chief Inspector of the Ministry of Natural Resources and Environment decided sanctioning of administrative violations of environmental protection for the total fine 267.5 million, forced to retrospectively pay environmental protection fees more than 127 billion. Besides, Vedan banned liquid discharge substandard Vietnam environment to environment. VEDAN paid compensation for damage costs, costs remedial environmental pollution of the Thi Vai River, compensation for economic losses, environmental for individuals and organizations affected for breach of VEDAN caused accordance with law.

-7th, October, 2008, Minister of Natural Resources - Environment Pham Khoi Nguyen has been officially reported to the Prime Minister on the test results. The same day, Minister Pham Khoi Nguyen also send documents requested provincial People's Committee issued a decision to suspend the production of Vedan Vietnam.

-13th, October, 2008, Prime Minister of Vietnam is directing the relevant authorities to urgently and resolutely implementing measures to handle violations of law on environmental protection for Vedan.

-5th, November, 2008, Ministry of Natural Resources -, the People's Committee of Dong Nai Province, Ltd. Vedan met on some contents of two decisions of administrative sanctions Inspectorate and the Ministry of Natural Resources and Environment decided to suspend discharge for Vedan.

-24th, October, 2008 Vice Chairman of Dong Nai province, said the province had written to the Prime Minister about the problems in the handling of VEDAN violation, which is competent to problems in the decision to suspend the operation of the company.

2.1.3. The issue of compensation for damage:

-13th, April, 2009 Vedan offering 3 levels of provincial support farmers HCMC, Dong Nai and Ba Ria-Vung Tau 25 billion. Then province Vietnam Farmers' Union launched support level 569.5 billion.

-May, 2009 Vietnam Farmers' Association has written to the Ministry of Natural Resources and Environment to support farmers in gathering evidence, assess environmental damage caused by Vedan as a basis for compensation.

-10th, June, 2009 Vietnam Environment Administration required HCM City, Dong Nai and Ba Ria-Vung Tau directed the departments of damage statistic and requests Vedan for compensation.

-8th, July, 2009 Vietnam Environment Administration delivered to the Institute of Natural Resources and Environment "report on environmental impact assessment and economic losses, environmental losses because of the Vedan's behavior and businesses on the Thi Vai River.

-4th, December, 2009 Vedan complete charge payment. However farmers 3 provinces Dong Nai, Ba Ria - Vung Tau and Ho Chi Minh City do not agree with the level of compensation of Vedan Vietnam so they decide to sue to Vedan Viet Nam for damages. After negotiation Vedan has agreed level of compensation due to farmers' representatives made and Ministry of Natural Resources and Environment has identified.

- 9th, August, 2010 Ministry of Natural Resources and Environment announced the results of the meeting on settlement of compensation for damage caused by Vedan VN. Accordingly, in addition to the amount of compensation to farmers 3 provinces, Vedan VN also must pay more than 4.5 billion, including 1.5 billion for operating expenses for statistical surveys of the damage made by authorities HCM City, BRVT, Dong Nai; more than 3 billion cost for environmental resource Institute (under Vietnam National University HCMC) and the Institute of Chemistry (Vietnam Science and Technology Institute) conduct surveys and assessments and making environmental impact assessment report and economic losses, environmental losses by acts of Vedan VN.

- 17th, August, 2010: Vedan Vietnam has moved 50% of the 45.7 billion in compensation for farmers in Can Gio, Ho Chi Minh City and 500 million for authorities have done environmental surveys. Under the agreement, the remaining amount will be transferred before the date 14th, January, 2011.
- 11th, August, 2010: Vedan decision to sign compensation 119 581 203 000 VND. According to general manager Yang Kun Hsiang of Vedan, compensation meets the 100% requirement in Official Letter No. 4901 / Committee-CNN
- 21th, June, 2010 People's Committee of Dong Nai province for farmers to claim almost 120 billion (this figure based on calculations of the Institute of Environment and Natural Resources). Farmers Union of Ba Ria - Vung Tau is also expected, from 13-21th, September to begin paying for the people, after moving Vedan compensation amount is 26.8 billion phase 1 (50% in total 53.6 billion).

2.1.4. Reference

[1]Literature:“Vedan case - legal regulation and dispute resolution practices, compensation for damages” – Dinh Hai Ninh KT33F.

URL:<http://luanvan.net.vn/luan-van/de-tai-vu-viec-vedan-quy-dinh-phap-luat-va-thuc-tien-giai-quyet-tranh-chap-boi-thuong-thiet-hai-29387/>

2.2 Formosa Environmental Incident [1]

2.2.1 Formosa Ha Tinh Steel Corporation Overview

Formosa Ha Tinh Steel Corporation (FHS) first invested in Vietnam since 2008 with a total investment of nearly 10 billion USD (with a capacity over 10 million tons / year in the first phase) and is planning to increase its capital to 27 billion dollars (20 million tonnes / year in phase II).

Construction started on December, 2012

URL:<http://kinhdoanh.vnexpress.net/tin-tuc/doanh-nghiep/formosa-de-xuat-lap-dac-khu-kinh-te-gang-thep-vung-ang-3009064.html>

FHS registered business: manufacture and sale of cast iron and steel; port business; cement manufacturing; Business power plants and water treatment plants; construction, installation, operation and trading gas plants, sale of compressed air and liquid gases for industrial use such as oxygen, nitrogen, ...; manufacturing, processing, production, sale and export of related products the coke, tar, light oil, and real estate business.

Formosa is been awarded the largest incentives as corporate income tax is 10% (domestic enterprises is 22%), income tax exemption for four years and 50% of the tax payable in the next 9 years; tax-free import of machinery and equipment; The excessive incentives for Formosa have raised question marks about the economic expertise.

URL:<http://baodatviet.vn/kinh-te/doanh-nghiep/formosa-nhan-boi-thuong-va-uu-dai-kich-tran-3046917/>

2.2.2 Pollution's Name

- Formosa Ha Tinh

2.2.3 Time

- 06/04/2016

2.2.4 Place

- Iron and Steel Hung Nghiep Formosa Ha Tinh Co., Ltd.
Vung Ang Economic Zone, Ky Anh Town, Ha Tinh Province

2.2.5 Damage [1]

- Fish die off in Ha Tinh, Quang Binh, Thua Thien Hue, Quang Tri Province.
- Seafood of dead by the disaster is over 100 tonnes
- 24.400 Unemployed

Among them, directly in the field of fisheries is 14,770 thousand. The number of unemployed people in the fisheries industry increased as 5736 people, logistics services sector increased 1,015 persons, the aquaculture industry increased by 823 people.

Also, in the field of hotel and restaurant services, the number of unemployed rose to 692, while in the field of salt production is 428 people.

- GDP of Viet Nam Decrease 0.3%
- The unemployment rate in Quang Binh has increased by 1.1%

2.2.6 Cause [2]

- Design and operation of wastewater treatment systems don't meet requirements, so coke oven discharges phenols and cyanide in the steel production process.
- Phenol, cyanide combined with iron hydroxides forming a complex mixture forms (MIXEL), greater proportion of sea, according to ocean currents move in the direction north-south from Ha Tinh to Thua Thien-Hue is the cause of seafood and marine life die series, especially at the bottom. This complex mixture forms took away all the oxygen in the seawater where it passes through

2.2.6 Solution [3]

➤ Government

- Steering timely implementation of measures to support and stabilize the lives of people stricken areas; maintaining political security, social order and social safety; Initial damage assessment of economic, social and environmental.
- At the same time, has directed the ministries, scientific institutions, in a spirit of careful, scientific, accurate, objective and lawful, define, clarify the cause, the object causing the problem order processing solution

➤ Enterprise

- Publicly apologize to the Government and people of Vietnam because of the occurrence of a serious environmental incident.
- To compensate for economic damage to the people and to support career change; compensation for pollution treatment and rehabilitation of the marine environment in the 4 provinces of central Vietnam with a total amount equivalent to more than 11,500 billion VND (500 million USD).
- Fix the existing radical, limitation of waste treatment systems, water treatment, improvement of production technology, ensuring thorough handling of hazardous waste before it is discharged into the environment to meet demand management of agencies of the central government and the province of Ha Tinh to not recur environmental incidents as happened.
- To coordinate with the ministries of Vietnam and the central provinces of building synchronized solutions to control the marine environment Central ensure prevention of pollution, not to environmental incidents similar order to build trust with the people of Vietnam and internationally.
- Correctly and fully implement the commitments referred to above, not to recurrence of violations of the law on environmental protection and water resources; if the violation will incur sanctions prescribed by the law of Vietnam.

2.2.7 Current Status[4]

- Observation assess the status, progress and extent of pollution of the marine environment and marine ecosystems and coastal four central provinces have applied the processes, methods based on science, in accordance Vietnam's regulations and relevant international.
- Regarding the quality of the environment and ecosystems:
 - a) The quality of seawater and marine sediments in the region have been observed within the limits prescribed by QCVN 10-MT: 2015 / BTNMT, to set standards for the beaches, water sports, aquaculture and aquatic conservation.
 - b) At 1.5 km from the coast area of Son Duong - Ha Tinh (area of about 300 square kilometer), door Nhat Le - Quang Binh (area 330 square kilometer), Hon Son Cha - Thua Thien - Hue (area of approximately 160 square kilometer), due to the impact of local eddy, there are some environmental parameters higher than other areas, but still within the permitted limits, should continue to be monitored and supervised closely and regularly.
 - c) With the strict control of emission sources from Hung Industrial Co., Ltd. Formosa Ha Tinh and by the natural cleansing mechanisms of the environment, concentrations of contaminants from environmental incidents tend to decrease with time.
 - d) Ecosystem reef, seagrass and marine resources in the area after the effects of environmental incidents severely degraded biodiversity both in scale and now has begun to show signs of recovery actively serving.
- About quality seafood caught: According to data of the Ministry of Health surveillance, from 04/28/2016 to 08/08/2016, the test results to assess the safety of seafood samples taken at 4 central provinces showed some levels of contaminants in seafood has decreased with time.

In the coming time, the Ministry of Health will coordinate with the Ministry of Agriculture and Rural Development to supervise seafood caught in the seas safe was Natural Resources and Environment Ministry announced.

2.2.8 Photos



Fish die off in 4 province of central

Reference

[1]B.B.H (12/2016), Environmental incidents Formosa as 24 449 people out of work, 0.3% of GDP damage, Revised at 07/02/2017, URL: <http://moitruong24h.vn/su-co-moi-truong-formosa-lam-24449-nguoi-mat-viec-gay-thiet-hai-03-gdp.html>

[2][3]Minh Ha, Dong Son (30/06/2016), Announce the cause of the incident marine environment abnormal death in 4 central coastal province, Revised at 07/02/2017, URL: <http://www.qdnd.vn/xa-hoi/cac-van-de/cong-bo-nguyen-nhan-su-co-moi-truong-lam-hai-san-chet-bat-thuong-tai-4-tinh-ven-bien-mien-trung-482056>

[4] MONRE (22/08/2016), Press information about the survey results, assess the extent and scope of the marine environment from pollution and environmental incidents caused at 4 central coastal province, Revised at 20/2/2017, URL: <http://botruong.monre.gov.vn/821/TNMT/2938/Thong-tin-bao-chi-ve-ket-qua-dieu-tra-danh-gia-muc-do-pham-vi-o-nhiem-moi-truong-bien-do-su-co-moi-truong-gay-ra-tai-4-tinh-ven-bien-mien-Trung.aspx>

2.3 Buoi river pollution in 2016

2.3.1. Case progress [1]

- 5, May, 2016 bright day, Le Duy Duong, Chairman of the People's Committees, Thach Thanh District , Thanh Hoa province , said the past few days, the Buoi river section running through the commune appeared mass death of fish situation undefined reason. As reported by the CPC Thach Lam commune, in 4 May 2016, the authorities receiving information from the Bien village, Doi village said along the Buoi river flows from the Hoa Binh province dead fish appeared on the river, green river turbid water, foul smell and white foam. People here asserted that polluted water sources causing mass death is due to agro-processing factory in Tan My commune, Lac Son district, Hoa Binh province discharged into rivers.
- In the morning of 07 May 2016, Director of Environmental Protection (Thanh Hoa province Department of Natural Resources and Environment), said that leaders of Hoa Binh Sugarcane Joint Stock Company (Based in the Tan My commune , Lac Son , Hoa Binh province) admitted to discharge untreated waste water into the Buoi river. The dirty discharge took place several days in a row during the second half of April to early May 5. This is the cause of mass death of fish in the river. In the afternoon of 6 may, in addition to the fish living in the natural environment, has nearly 7 tons of fish in cages on the Buoi river at Thach Lam Quang Thach , Thach Cam , Thanh My commune (Thach Thanh district) died. Up to 10:00 o' clock in 7 May, the total number of dead fish is 17,385 kg, including 73/109 and 32/49 cage fish farms completely dead.
- According to Dan Tri newspaper, the Ministry of Natural Resources and Environment has directed Vietnam Environment Administration to establish working group with the People's Committee of Thanh Hoa province. Missions of the Ministry of Natural Resources and Environment will be responsible for supporting Thanh Hoa to determine the cause of the pollution, pollution levels and measures to redress the environmental pollution on Buoi River. Simultaneously taking measures to handle violations of JSC Sugar Hoa Binh as a basis to ask the company to compensate the economic damage to the people.
- In 14 May 2016 more fish cages of households on the Buoi river died in Cam Thach commune, Thach Dinh commune. In the evening of 15 May 2016, 7 households at Cam Thach,

3 households at Thach Dinh dead fish cages with a total weight of more than 1.1 tons. The majority of dead fish are carp, came to maturity, the weight of 2-4 kg / head.

2.3.2 The solution of authorities and the results [2]

- *The General Director of the General Department of the Environment has issued a decision to sanction 3 companies and individuals a total of 3.9 billion by the major polluters on the Buoi River (Thanh Hoa) causing mass death.*
- In 17, May, 2016, *The General Director of the General Department of the Environment* (Ministry of Natural Resources and Environment) issued the Decision sanctions against organizations and individuals that discharge activities causing pollution environment in the Buoi river with a total fine of up to 3,904,165,200 VND contracts, along with the remedial measures attached.
- Earlier, implement the directive of the Minister of Natural Resources and Environment Tran Hong Ha dated 10-5-2016 of the inspection of the observance of laws on environmental protection for the facility in the Tan My commune, Lac Son district, Hoa Binh province, the inspection team has conducted inspections to 3 basis are: Tan Hieu Hung Limited company, Hoa Binh sugar JSC and pig breeding establishments Nguyen Ngoc Sang (by Nguyen Ngoc Sang-employed). This is the basis of operation to discharge pollutants in the river, killing fish in the Buoi River recently.
- Based on the results of the inspection team, dated 17-5-2016, *The General Director of the General Department of the Environment* issued the decision sanctioning of administrative violations in the field of environmental protection for organizations and individuals human above, including:
 - + Sanctions against Tan Hieu Hung Limited Company total amount of 1,926,666,300 VND for acts such as: Failing to properly and fully content and requirements in conclusion, inspection of environmental protection investigations of state agencies authorized; Improper performance of content requirements, progress of the environmental pollution treatment; Not implementing monitoring programs, environmental monitoring prescribed; Improper performance of the contents of a report of environmental impact assessment has been approved ...
 - + Sanctions against Hoa Binh JSC Sugar a total amount of 1,783,332,600 VND for acts such as not formulated and approved environmental management plan as prescribed; Not implementing monitoring programs, environmental monitoring prescribed; No construction works for

environmental protection according to regulations (for the waste water treatment system); Wastewater discharge exceeded technical regulations on waste; ...

+ Sanctions against Mr. Nguyen Ngoc Sang total amount of 194,166,300 VND shall be imposed for acts such as not implementing monitoring programs, environmental monitoring prescribed; Wastewater discharge exceeded technical regulations on waste; ...

+ In addition, the facility will be required to suspend the operation of wastewater discharge environmental pollutants; suspension of production operations and to immediately implement the remedial measures required by the inspection team.

- In 17, May, 2016, the Department of Agriculture and Rural Development (DARD) in Thanh Hoa province has reported test results of water samples and dead fish in the Buoi River, running through Thach Thanh district, Thanh Hoa province.
- Accordingly, based on the environment law, exceeding the limitation of NH_3 is 7.4 to 7.5 times, H_2S exceeds allowable limit is 1.5-2 times, norms NO_2 exceeds allowable limit is 2-4 times. Disease, clinical examination on the fish, found no signs of infectious disease. From the above results, Thanh Hoa DARD confirmed dead fish in the Buoi River by environmental factors heavily contaminated.
- According to Thanh Hoa DARD, solutions to overcome the water by means of biological and chemical are not feasible because the volume of contaminated water is too large. Therefore, water needs time for flood running off and self-cleaning ability of the natural environment. In the report also showed that the source of contaminated water has affected the livestock sector, particularly farming and life of the people living along the Buoi River. Currently, DARD is directing for Thach Thanh District People's Committee and relevant departments take measures to limit the damage to downstream (when water pollution not coming yet) as moving cage, harvested fish, fish move into the pond, recommending the resident living along the river should not use the Buoi river, especially for use in daily life.
- Earlier, as *Nguoi Lao Dong* Newspaper has announced, starting from the morning of 4, May, the Buoi river, flows through Thach Lam (Thach Thanh district, Thanh Hoa province) appear **mass death of fish** situation, water turn blue opaque and foul smell. The following day, the water on the downstream drift caused mass death of fish cages in Thach Quang, Thanh Vinh, Thach Thanh district....

- Shortly after the fish dies in 5, May, Thanh Hoa province Department of Natural Resources and Environment in coordination with the Hoa Binh province Department of Natural Resources and Environment established the working group inspecting the discharge of the factory in Hoa Binh province, including Hoa Binh sugar Joint Stock company (in Tan My commune, Lac Son district, Hoa Binh province). Sugar Factory then admitted the error - discharging untreated water into Buoi Rivers leading to a series of dead fish, polluted river water.
- Hoa Binh JSC sugar then agreed compensation, assistance for 34 households damaged by wastewater 1 billion 404 million (with the number of dead fish was 17.5 tons, the price of 80,000 VND / kg).
- Related to the discharge of wastewater into the environment, also during the day 17 May 2016, General Department of Environment (MONRE) has a decision to sanction Hoa Binh JSC sugarcane 1.783 billion and suspend operations 6 months beginning on the date of 20 May 2016. Also, Request the company to establish the basic design for the construction of environmental protection under the provisions and to urgently build wastewater treatment plant before discharging wastewater into the Buoi river drainage ditch.

2.3.3 Reference

URL:

[1]https://vi.wikipedia.org/wiki/S%C3%B4ng_B%C6%B0%E1%BB%9F#M.C3.B4i_tr.C6.B0.E1.BB.9Dng

[2]<http://nld.com.vn/thoi-su-trong-nuoc/phat-39-ti-dong-thu-pham-gay-o-nhiem-khien-ca-chet-hang-loat-20160517230318592.htm>

2.4. Environmental pollution of the Nui Phao project

2.4.1 Case progress [1]

- Do not deny the positive jobs - Nui Phao Mining Mineral Processing did help to mitigate the negative effects of mining polymetallic mine recently. However, due to unrealized so radically since officially embarked in exploiting in 2013, people living in Ha Thuong commune, Dai Tu district, Thai Nguyen province has often had to carry the petition to functional branches and levels of government.

- From September 2015 to now, people living in Ha Thuong, Dai Tu district, response on the issue intensified, more acute, signs of agitation, affect social order and security in this area. Most recently, in 13 June 2016 - afternoon, several people in 4 area, Ha Thuong Commune, gathered and obstruct traffic, throwing stones down the road, carrying the coffin out to block the internal way of the Nui Phao Company. Due to the propaganda and mobilization of authorities, in 6 March 2016 - night people dispersed.
- Most of the comments of the people living in the areas reflected that three major pollution issues upsetting their lives are noise, air and water.
- Tran Thi Dung (Head Women's Association 3 area, Ha Thuong commune, Dai Tu district) said that the company has made improvements Highway 37 service by creating pass tens of meters higher than residential. In the morning and at night Vehicles moves noisily, discharge smoke and dust to neighbor. Moreover, the new Pass Road restricted traffic, make it difficult for economic development. Along the pass road, dust and noise of the factory effected the people living near here.
- Ms. Nguyen Thi Tuong (Head 4 area, Ha Thuong commune) said: 24/24 hours, people have heard the crushed ore, additional blasting ore in large quantities often shocked homes and land. Conveyor ore powder tens of meters high into the air exhaust dust flying around the region together with the odor causing people trouble breathing, vomiting.
- Especially, when the Nui Phao Company operate to be really the horror of households living near the factories area. Every day, every convoy carrying chemical tanks caught all kinds of smells that when people inhale causing sneezing, runny nose, chest tightness, shortness of breath, vomiting series. Implications of this situation is the sudden increase of respiratory diseases, skin diseases, rashes, itching, fighting off forever.
- There have been dozens of records on air pollution created in the 4 area. Particularly, in these days from 19 to 26 November 2015, there are 7 reports were signed and sealed by Mr. Tac Van Loi -Vice Chairman of Ha Thuong Commune.
- In addition to noise and air, the water discharged into the environment of the project also aching in the community. According to many people, black wastewater discharged, causing headaches, fatigue, mouth smell once saw bitter. Some households in the neighborhood did not dare to use water of families that have switched to buy a pure water. Waste water spilled on the natural

environment are also factors that cause crop yield loss. Even when some farmers stepped into the wastewater which caused limb ulcers.

2.4.2 Reaction from the government [1]

- Concerning with this problem, Nguyen Van Hong, chairman Ha Thuong Commune People's Committee, said: in 2012, Thai Nguyen province People's Committee has advocated for compensation Nui Phao Company, support the people's movement around 3, 4 area Ha Thuong commune from the area near the plant. In August 2014, Thai Nguyen Province People's Committee making land acquisition policy of the environment affected households. However, the implementation process is very slow, nearly 5 years - a resettlement site unfinished, additional land acquisition plan of 41 households directly affected have not been implemented.
- According to Chairman of the Ha Thuong commune People's Committee, where environmental issues have a direct impact on the lives of the people close to the region of mining and manufacturing enterprises. Meanwhile the wastewater treatment system by Nui Phao processed only in the process of deposition over the reservoir and then discharged into the environment.
- According to the results of environmental monitoring by the Thai Nguyen Department of Natural Resources and Environment hosted shows that the wastewater of Nui Phao company discharging into in Cat stream (source receiving wastewater) were contaminated with heavy metals and chemicals tight. Levels of arsenic, iron, mercury, fluorine, total Cyanide has exceeded permissible limits.
- Due to the reflection of regional people, especially people of Ha Thuong commune 3.4 area, Dai Tu district; in recent years, Thai Nguyen People's Committee has regularly directed the Department of Natural Resources and Environment, Dai Tu District People's Committee and relevant agencies actively, proactively solving competence of environmental issues in Nui Phao project.
- To solve some existing problems in the work of the clearance of Nui Phao project, in 24 April 2014, Thai Nguyen Provincial People's Committee has Notice No. 41 / TB-UBND which has agreed land recovery policy at 3,4 areas, Ha Thuong village to serve the Nui Phao project; required the company to give a plan of the Nui Phao project adjustments and additional land acquisition for 3.4 area Ha Thuong village, as basis for land acquisition under the provisions.

- On the other hand, Thai Nguyen Province People's Committee also regularly report to the Ministry of Natural Resources and Environment on the operations of the Company and the impact of the Nui Phao on the environment, regional people's lives; The Ministry has been proposed to give the measures to solve completely environmental problems caused by Nui Phao Company.
- Particularly In June of 2016, the People's Committee of Thai Nguyen province has issued 03 documents (No. 915/03.31.2016 UBND-KTN; No. 1441/UBND-CNN dated 05.10.2016, No. 1968/UBND-CNN days 06.15.2016) Reporting Ministry of Natural Resources and environment on the operations of the company and the impact of the Nui Phao affecting the environment, regional people's lives; request the Ministry of Natural Resources and Environment to inspect, evaluate the overall impact on the environment from the operation of the company to take measures Nui Phao final resolution on the environmental issues of the Company Nui Phao.
- In Report No. 147 / BC-UBND dated July 2016 and in meetings with the Thai Nguyen Ministry of Natural Resources and Environment. In 14 July, Thai Nguyen province have suggested the Ministry of Natural Resources and Environment comprehensive inspection of the environmental protection maintenance in mining, mineral processing by Nui Phao company, as the basis for the settlement, answer people's letters of recommendations on the issue of environmental pollution in the region.
- The document of Thai Nguyen Province People's Committee, said: "In the production process, the Nui Phao Company has implemented the environmental protection measures such as wastewater treatment before discharging into the environment; spraying water to remove dust on the internal roads of the mine; spraying moisture to reduce dust and noise resistance for conveyor belts; additional dust screens; construction of wastewater treatment system, by the end of 2015 has been completed (trial operation since January of 2016); monitoring and reporting the full results of environmental monitoring periodically, ... After all, production activities of the Nui Phao company still cause impact to the surrounding environment and people's lives in residents's areas.
- From the word "cry" of the people and government at all levels in Thai Nguyen, the Ministry of Natural Resources and Environment has planned to inspect the problems of implementing

regulations and policies of the environment laws on mining at the Nui Phao project within august of 2016

2.4.3 Results [2]

- Representing the Ministry of Natural Resources and Environment has updated the initial results at Nui Phao project.
- At a news conference in the morning of 7 november, Inspector MONRE updated inspection results initially at 3 **Nui Phao** (Thai Nguyen). According to Hoang Van Thuc, deputy general director of the General Department of the Environment, the project are subject to **unannounced inspections** under the direction of leaders of the Ministry of Natural Resources and Environment, the information from the news and local requirements. Now, although there are no final conclusions of the inspection, initially but has also found a number of violations.
- Nui Phao project, representatives Vietnam Environment Administration confirmed that the official conclusions will be made public. It is known that inspection activities of Nui Phao mineral processing company (Thai Nguyen) is starting from 28 september 2016, Inspection period is 45 working days, excluding weekends and holidays so far has not ended. "The inspection team Nui Phao do not have any public obstruction. Enterprises are also very open to report relevant information," said by Hoang Van Thuc.

2.4.4 Reference

URL:

[1]<http://hoanhap.vn/bai-viet/o-nhiem-moi-truong-tai-du-an-nui-phao--keu-cuu-tu-thai-nguyen-11887>

[2]<http://www.baogiaothong.vn/ket-qua-ban-dau-thanh-tra-vu-nui-phao-leeman-va-dap-dinh-vu-d176836.html>

2.5 Hapaco Hai Au Limited Company discharges wastewater into Lach Tray river without treatment in Hai Phong city

Time: From many years, people constantly reflecting on Hapaco Hai Au Co., Ltd (production of paper) continuous discharge of wastewater into Lach Tray river, wastewater is yellow, sometimes black, fishy odor concentration

Location: residential areas Ton Duc Thang, An Dong Commune, An Duong district, Hai Phong

7th July, 2016 Department of Natural Resources and Environment of Hai Phong with local authorities to inspect the wastewater of Hapaco Hai Au Co., Ltd. Inspection team has discovered a series of violations of this company.

Reason: This company is discharging waste water into the environment without discharge permits; Waste water treatment system of the company by iron, installed since 2002 was rusty, damaged, severely deteriorated and are not maintained, operated regularly, not enough wastewater treatment capacity; wastewater treatment system is not operating properly process.

This company don't carry sludge treatment that according to the law.

Solution: Inspection team has asked the company to stop the waste discharge.

20th August, 2016 Hai Phong City People's Committee has decided to sanction the company for violating the environmental field with the amount of over 400 million VND.

Hapaco Hai Au Co., Ltd admits there are violations, shortcomings in the management and treatment of waste water in paper production. So, some indicators wastewater exceeds permitted limits.

Reasons explained by perennial infrastructure, old technology devices, often have problems the device should not be sufficient response environmental requirements. 22th August, 2016 Hapaco Hai Au Co., Ltd announced decision to stop production and relocation of enterprises related to the paper for violations of environmental protection.

Reference

URL:

<http://www.vtc.vn/nha-may-giay-gay-o-nhiem-song-lach-tray-bi-phat-400-trieu-dong-dung-san-xuat-d272461.html>

<http://laodong.com.vn/xa-hoi/nha-may-giay-gay-o-nhiem-moi-truong-tai-hai-phong-yeu-cau-dung-hoat-dong-xa-thai-579491.bld>

2.6 Copper factory A Cuong Group waste discharge into Cam Dan river without treatment in Bac Giang province

Time: From many years, this factory continuity had been discharged wastewater without treatment into Cam Dan river.

Location: Cam Dan commune, Son Dong district, Bac Giang province.

Only from 2014 onwards, this factory have been the authorities in Bac Giang province several times prepare minutes and administrative sanctions for acts of discharging wastewater directly into Cam Dan river without treatment.

Most recently, on 26/6/2016, interdisciplinary inspection team from the Department of Natural Resources and Environment of Bac Giang province to lead and coordinate with provincial police PC49 room has examined the actual implementation of the law on environmental protection of the company and discovered that waste water containing mud and dark opaque untreated prescribed and discharge directly to Cam Dan river.

Reason: copper selection factory with capacity of 1,000 tons/year has direct discharge of waste water untreated into the environment; not yet fully implemented content Environmental Protection Schemes, not build wastewater recirculation system.

Solution: Inspection team have requested the company to stop operations discharging wastewater containing sewage sludge has not been collected and handled according to regulations to the environment.

Make plan to solution treatment for dredging sludge deposited on Cam Dan River, and report to Department of Natural Resources and Environment of Bac Giang province.

8th July, 2016. Bac Giang Province People's Committee has asked the company to stop operations discharging wastewater directly into the environment.

Reference

URL:

<http://www.baotainguyenmoitruong.vn/ban-doc/201607/cong-ty-a-cuong-ngang-nhien-xa-thai-trai-phep-dau-hieu-bat-thuong-trong-quan-ly-2712690/>

<http://giadinhphapluat.vn/bo-truong-chi-dao-thanh-tra-cong-ty-xa-thai-ra-song-o-bac-giang-p38371.html>

2.7 CKC Limited Company with the waste reservoir subsidence incidents of lead the floatation plant in Cao Bang Province.

a. Company overview

CKC Company was established since 2009, CKC is the investor of the project processing plants lead – zinc in Bao Lam district, Cao Bang Province.

The plant has a capacity of 20,000 tons of lead - zinc metal / year.

From July 2011, processing plants lead - zinc officially put into operation. However, the operation is not continuous and does not reach the designed capacity because of unstable power supply and input source of lead – zinc is not enough.

October 2014, the plant is temporarily shut down, and continue to produce again from November 2015.

b. Environmental incident

Time: At 14:30pm 5th January, 2016. Incident broken sewer at starting point of waste water drainage system of sludge reservoir.

At 5:00 am 6th January, 2016. Incidents continued of subsidence on the dry sand at the waste reservoir near the dam.

Location: Lang Ca village, Pac Miao town, Bao Lam district, Cao Bang Province.

Reason: The total volume of waste discharged into the environment about 2,000 m³ of waste water and sewage sludge. Flowed directly into Gam river and flowed on the surface of the resident's arable land with area of about 1,000 square meter, not damage to human life.

Reference

URL: <http://tnmtcaobang.gov.vn/index.php?language=vi&nv=news&op=Moi-truong/Cong-ty-TNHH-CKC-va-su-co-sut-lun-ho-chua-thai-cua-nha-may-tuyen-noi-chi-kem-tai-xom-Lang-Ca-thi-tran-Pac-Mieu-huyen-Bao-Lam-2318>

3. Good Protection Environmental Activities in Provinces

3.1 Binh Duong Province

- Binh Duong province release Binh Duong Green Book to honor, reward incentives, arousing the spirit of enterprise pride in maintaining good environmental protection. Named to Binh Duong Green Book in 2016, the trademark of the company and the information will be widely promoted. Not only Binh Duong Green Book also helps enterprises to integrating better with the world trade market. Therefore, named to Binh Duong Green Book is a safest stepping stone for businesses when exporting goods, as well as affirm the brand name of the enterprises, company's products against consumers.
- URL: <http://baobinhduong.vn/sach-xanh-binh-duong-ton-vinh-doanh-nghiep-bao-ve-moi-truong-a152391.html>

3.2 Ba Ria – Vung Tau Province

- Department of Natural Resources and Environment has proposed Ba Ria - Vung Tau People's Committee directing Dat Do district, Xuyen Moc district People's committee concentrate investment 02 seafood concentrate processing clusters in Loc An and Binh Chau. Since 2017, the organization moved the seafood processing facilities that caused pollutants into seafood concentrate processing clusters
- URL: <http://lvsdongnai.cem.gov.vn/b%C3%A0-r%E1%BB%8Ba-v%C5%A9ng-t%C3%A0i-nhi%E1%BB%81u-v%E1%BA%A5n-%C4%91%E1%BB%81-m%C3%B4i-tr%C6%B0%E1%BB%9Dng-c%E1%BA%A7n-%C4%91%C6%B0%E1%BB%A3c-x%E1%BB%AD-l%C3%BD.aspx>

3.3 Long An Province

- Green Environment Technology Park Project
- In the morning of 9-November-2014, the investor California Waste Solutions, Inc., (CWS) and the Vietnam Waste Solutions, Inc. (VWS) has done ground-breaking ceremony to build bridges, pave the way to Green Environment Technology Park project in Tan Lap commune, Thu Thua district, Long An province.
- Green Environment Technology Park covers an area of 1,760 hectares, the design capacity of 40,000 tons of waste / day, the first phase investment of 500 million USD and the operating time of the project from 75-100 years.
- This is a complete waste disposal complex, including construction items such as: unloading wharf area, receiving waste, recyclable materials, building materials, goods and containers; high-tech domestic waste disposal area; medical waste disposal area; recycling plants including versatile classification recycling plant, compost processing workshops, sludge treatment plants, domestic wastewater and rain water treatment plants, treatment plant of the leachate water from burial sites high tech area; reception area of e-waste and hazardous waste from households; renewable plant waste energy from burning the waste after sorting ... will increasing. Therefore Green Environment Technology Park is designed to dynamic combine with the waste separation process at different places. This feature helps project good response according to changing waste composition rates in the future.
- The Long An Green Environment Technology Park are applied waste treatment technologies most advanced of the world, such as renewable energy technologies from waste, gasification, the process of regeneration, recycle materials ...
- The project will dedicate a separate area for types of waste to processing, recycling and sorting electronic waste (E-waste), hazardous waste from households as those needs arise in the future.
- URL:<http://nld.com.vn/kinh-te/khoi-dong-du-an-khu-cong-nghe-moi-truong-xanh-20141111183804738.htm>

3.5 Dong Nai Province

Dong Nai province conduct controlling the way of hazardous waste, solid waste

- Department of Natural Resources and Environment of Dong Nai Province have built cruise monitoring system for hazardous waste transport and management of solid waste in order to closely manage the implementation of the provisions of law in the management of solid waste and contribute to the management of solid waste in a scientific manner, progressive and timely; especially in the course of collection, transport and disposal of hazardous waste an online, automated and continues.
- URL:<http://stnmt.dongnai.gov.vn/Pages/pTinTucChiTiet.aspx?newsid=6554>.

3.6 Ho Chi Minh City

3.6.1 Experiment organic waste easily biodegradable treatment plant renewable energy generators.

- HCM City People's Committee has agreed Hitachi Zosen Corporation Company (Hitz) and Company Limited City Urban Environment (CITENCO) co-operate implemented "test organic waste easily biodegradable treatment plant renewable energy generators "with a capacity of 200 kg / day of food waste is collected from the program classification of solid waste at source, Ben Nghe ward, district 1.
- The location of experiment model belongs to Phuoc Hiep landfill site management by CITENCO (in the complex treatment of solid waste Northwest Cu Chi District), an area of about 300 m2, on the entire facility to gain environmental standards, fire protection according to regulations. Hitz and CITENCO committed to properly and fully implement the environmental protection measures in accordance with the law of Vietnam and return the cleaned surface after the end of the test time; and pledged to cooperate to carry out test model is not related to the implementation of future projects (if applicable). Investment funding model building, machinery and equipment (100% guaranteed) and operational management, food waste transportation after sorting from a centralized location to location installation test model due to Hitz of payment. Investment funding model building, machinery and equipment to ensure 100% new and operational management, food waste transportation after sorting from a centralized location to location model installation testing expenditure by Hitz of payment. Time trial 8 months from October.
- URL:<http://www.donre.hochiminhcity.gov.vn/tintuc/Lists/Posts/Post.aspx?List=f73cebc3-9669-400e-b5fd-9e63a89949f0&ID=6636>

3.6.2 Ho Chi Minh City Environmental Sanitation Project

- The Ho Chi Minh City Environmental Sanitation (Nhieu Loc-Thi Nghe Basin) Project is having a transformational impact on the city, directly benefitting over 1.2 million people with improved sanitation and reduced flooding while at the same time serving as a new city asset that can be enjoyed by the people of Ho Chi Minh City.

Challenge

- Development challenges that the Project tackled included: (a) insufficient capital investment to replace decaying infrastructure and to keep pace with rapid urban growth; (b) the frequent flooding that occurred in cities because of undersized drains and inadequate routine maintenance, (c) severe pollution of water courses, particularly in densely populated areas of cities, because there was no provision for wastewater collection and treatment; and (d) weak utility agencies.
- Implementation of the project had also been challenging because:

- The project was technically complex, involving tunneling works which were often subject to unpredictable surface soil and groundwater conditions.
- The client was unfamiliar with the engineering technology and with Bank procurement procedures, repeated changes in the project management, high inflation costs or problematic performance of several contractors.
- The project site was located in the most densely populated central business district of the city with critical traffic management issues.

Solution

- For 20 years, Nhieu Loc – Thi Nghe was a dark and smelly canal full of waste. Begun in 2002, the project was designed to address these issues. This involved the construction of a sewer interceptor under the canal, 8 kilometers long and 3 meters in diameter, to improve the canal's drainage capacity, as well as strengthening of the canal's embankments and installing over 60 kilometers of sewer lines.
- Because the project is located in the central business district, construction work could only be done at night for the most part. The unpredictable relocation of existing underground water pipes – including a 2 meter diameter water main, power and telecommunication cables added to the challenge. The project applied tunneling works in Vietnam's unpredictable sub-surface soil and groundwater conditions that had never been done before.

Results

- The Project has had a transformational impact on the city, lifting the city's urban face. Thanks to the Project, 96,000 households (400,000 people) now enjoy reduced flooding, and 240,000 households (1.2 million people), mostly poor, now have centralized wastewater collection. Fish are now returning to the Nhieu Loc Thi Nghe canal, proving that the water quality in the canal has been significantly improved.
- Between 2002 and 2012, the project helped to install a 9 km wastewater interceptor with an inside diameter of 2.5-3.0m with a pipe-jacking technology that has been applied for the first time in Vietnam. It also replaced and extended: i) 51km of combined primary and secondary sewers; and ii) 375km of tertiary sewers.
- It also conducted: i) dredging, transportation and disposal of approximately 1.05 million m³ of sludge and excavated material to increase the hydraulic capacity of the canal; and ii) strengthening of 18km of canal embankments by concrete sheet pipe.

Moving Forward

- Phase II of the project is under preparation. It includes: i) a wastewater treatment plant; ii) remaining interceptor; and iii) sewerage in District 2. This will be the natural continuation of the project as it will treat the wastewater that is now being temporarily discharged into the Saigon

river. With the preliminary cost estimate of US\$478 million, the Board date is expected in December. 2013.

URL:<http://www.worldbank.org/en/results/2013/04/11/vietnam-ho-chi-minh-city-environmental-sanitation-project>